Exploratory Data Analysis

```r
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.1     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```r
sv <- read_csv("data/clean/g24_svprec_clean.csv")
```

```
Rows: 51123 Columns: 76
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (29): FIPS, SVPREC, SVPREC_KEY, ELECTION, GEO_TYPE, ASSAIP01,
ASSDEM01, ...
dbl (47): COUNTY, ADDIST, CDDIST, SDDIST, BEDIST, TOTREG, DEMREG, REPREG,
AI...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

Distribution of total votes per precinct

```r
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```
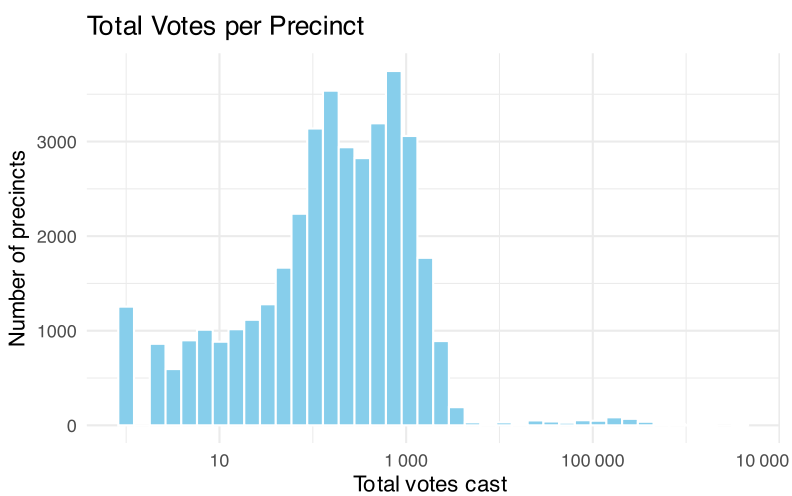

```
sv_with_votes <- sv %>%
  filter(TOTVOTE > 0)
sv_precinct_totals <- sv %>%
  mutate(total_votes_precinct = rowSums(across(ends_with("VOTE")), na.rm =
TRUE))
ggplot(sv_precinct_totals, aes(x = total_votes_precinct)) +
  geom_histogram(bins = 40, fill = "skyblue", color = "white") +
  scale_x_log10(labels = scales::label_number()) +
  labs( title = "Total Votes per Precinct", x = "Total votes cast ", y =
"Number of precincts") +
  theme_minimal()
```

```
Warning in scale_x_log10(labels = scales::label_number()): log-10
transformation introduced infinite values.
```

```
Warning: Removed 12466 rows containing non-finite outside the scale range
(`stat_bin()`).
```

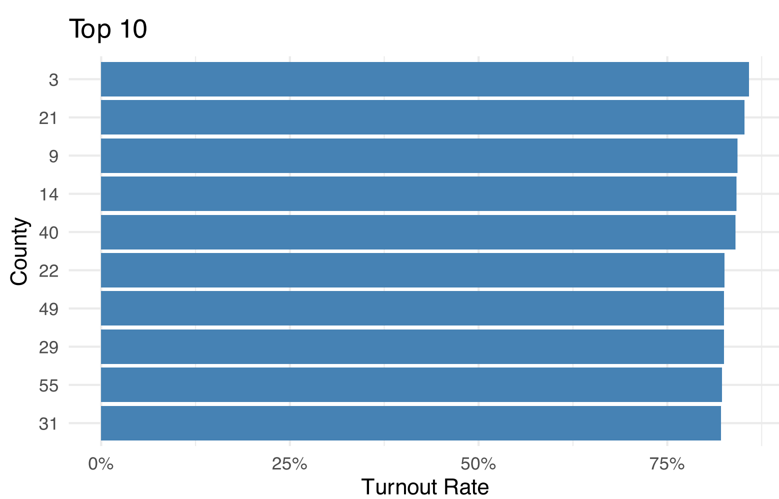


Answer 1

Based on the data, there is a relativeley small number of pricincts with very few total votes cast (around 5), a large number of precincts with 750-1250 total votes cast some precincts, and very few precincts with ~100,000 total votes cast.

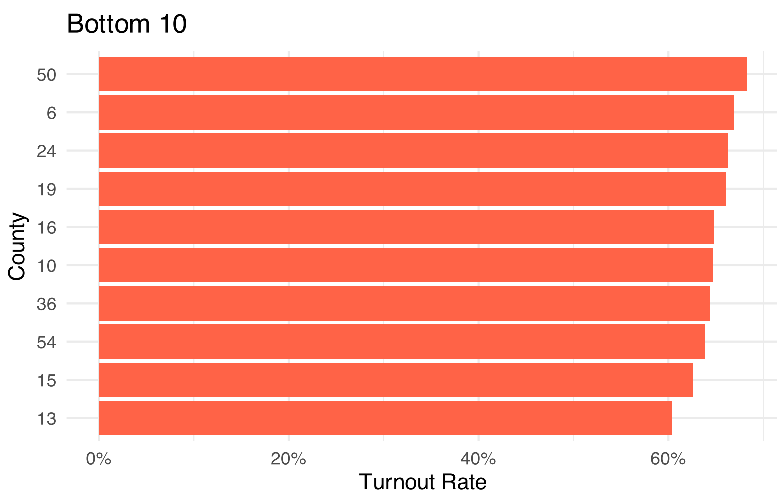
What were the Bottom/Top 10 Counties by Turnout Rate?

```
county_turnout <- sv %>%
  group_by(COUNTY) %>%
  summarise(turnout = sum(TOTVOTE, na.rm = TRUE) / sum(TOTREG, na.rm = TRUE))
%>%
  filter(!is.na(turnout))

# Top 10 plot
county_turnout %>%
  slice_max(order_by = turnout, n = 10) %>%
  ggplot(aes(x = reorder(COUNTY, turnout), y = turnout)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  scale_y_continuous(labels = scales::percent) +
  labs(
    title = "Top 10", y = "Turnout Rate", x = "County") +
  theme_minimal()
```




```
# Bottom 10 Plot
county_turnout %>%
  slice_min(order_by = turnout, n = 10) %>%
  ggplot(aes(x = reorder(COUNTY, turnout), y = turnout)) +
```

```
  geom_col(fill = "tomato") +
  coord_flip() +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Bottom 10", y = "Turnout Rate", x = "County") +
  theme_minimal()
```



Answer 2

Based on the two plot, County 13 had the lowest turnout rate and County 3 had the highest turnout rate.

How did winners look on a precinct-level?

```
library(tidyverse)
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i Use `spec()` to retrieve the full column specification for this data.
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```
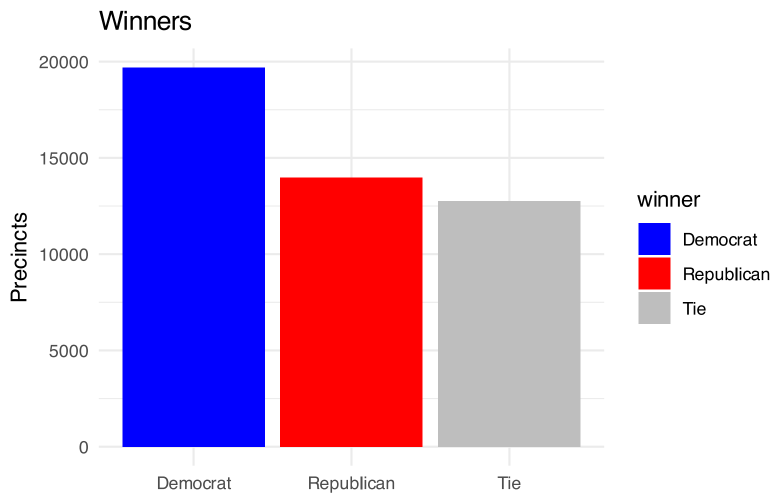

```
dem <- "CNGDEM01"
rep <- "CNGREP01"
sv_clean <- sv %>%
  mutate( DEM_CNG = parse_number(.data[[dem]]),REP_CNG =
parse_number(.data[[rep]])) %>%
  filter(!is.na(DEM_CNG) & !is.na(REP_CNG)) %>%
  mutate( winner = case_when (REP_CNG > DEM_CNG ~ "Republican", DEM_CNG >
REP_CNG ~ "Democrat", DEM_CNG == REP_CNG ~ "Tie"))
```

```
Warning: There were 2 warnings in `mutate()`.
The first warning was:
i In argument: `DEM_CNG = parse_number(.data[["CNGDEM01"]])`.
Caused by warning:
! 4684 parsing failures.
row col expected actual
 19  -- a number    ***
 71  -- a number    ***
147  -- a number    ***
179  -- a number    ***
213  -- a number    ***
... ... ........ ......
See problems(...) for more details.
i Run `dplyr::last_dplyr_warnings()` to see the 1 remaining warning.
```

```
sv_clean %>%
  count(winner) %>%
  ggplot(aes(x = winner, y = n, fill = winner)) +
  geom_col() +
  scale_fill_manual(values = c("Democrat" = "blue", "Republican" = "red",
"Tie" = "gray")) +
  labs( title = "Winners", x = NULL, y = "Precincts") +
  theme_minimal()
```



Answer 3

On a precinct-level, Democrats are the winner.